The parameters in optical systems interact in many unexpected ways. For example, consider how the amount of available light and the sensitivity of the light sensor affects the *sharpness* of the acquired image. This is because the *iris diameter* and the *exposure time* are adjusted to transfer the proper amount of light from the scene being viewed to the image sensor. If more than enough light is available, the diameter of the iris can be reduced, resulting in a greater *depth-of-field* (the range of distance from the camera where an object remains in focus). A greater depth-of-field provides a sharper image when objects are at various distances. In addition, an abundance of light allows the exposure time to be reduced, resulting in less blur from camera shaking and object motion. Optical systems are full of these kinds of trade-offs.

An adjustable iris is necessary in both the camera and eye because the range of light intensities in the environment is much larger than can be directly handled by the light sensors. For example, the difference in light intensities between sunlight and moonlight is about one-million. Adding to this that reflectance can vary between 1% and 99%, results in a light intensity range of almost *one-hundred million*.

The **dynamic range** of an electronic camera is typically 300 to 1000, defined as the largest signal that can be measured, divided by the inherent noise of the device. Put another way, the maximum signal produced is 1 volt, and the rms noise in the dark is about 1 millivolt. Typical camera lenses have an iris that change the area of the light opening by a factor of about 300. This results in a typical electronic camera having a dynamic range of a few hundred thousand. Clearly, the same camera and lens assembly used in bright sunlight will be useless on a dark night.

In comparison, the eye operates over a dynamic range that nearly covers the large environmental variations. Surprisingly, the iris is not the main way that this tremendous dynamic range is achieved. From dark to light, the area of the pupil only changes by a factor of about 20. The light detecting nerve cells gradually adjust their sensitivity to handle the remaining dynamic range. For instance, it takes several minutes for your eyes to adjust to the low light after walking into a dark movie theater.

One way that DSP can improve images is by reducing the dynamic range an observer is required to view. That is, we do not want very light and very dark areas in the same image. A reflection image is formed from *two* image signals: the two-dimensional pattern of how the scene is *illuminated*, multiplied by the two-dimensional pattern of *reflectance* in the scene. The pattern of reflectance has a dynamic range of less than 100, because all ordinary materials reflect between 1% and 99% of the incident light. This is where most of the *image information* is contained, such as where objects are located in the scene and what their surface characteristics are. In comparison, the illumination signal depends on the light sources around the objects, but not on the objects themselves. The illumination signal can have a dynamic range of millions, although 10 to 100 is more typical within a single image. The illumination signal carries little interesting information,